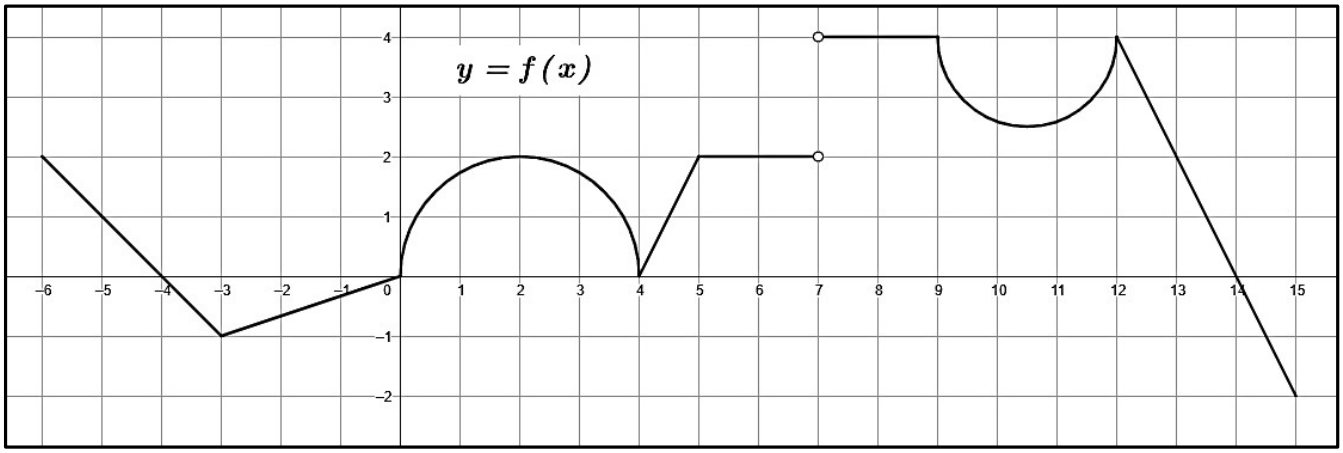
Worksheet 6-7 A
The Fundamental Thm of Calculus Part 1 #1
AP Calculus Name_____



The graph of $f(x)$ is above and consists of six linear pieces and two semicircles. Use the graph to find the following:

1.) $\int_{-6}^{0} f'(x)\,dx$

2.) $\int_{4}^{6} f'(x)dx$

3.) $\int_{13}^{14} f''(x)dx$

4.) $\int_{0}^{7} f(x)\,dx$

5.) $\int_{14}^{9} f(x)dx$

6.) $\int_{-5}^{0} [2 - f(x)]dx$

7.) $\int_{-5}^{15} f'(x)\,dx$

8.) $\int_{13}^{15} [f'(x) + 2]dx$

9.) $\int_{-2}^{-4} [f'(x) + 3]dx$

The function $f(x)$ is continuous and three times differentiable. Selected values for F, f, f' and f'' are given in the table below. Use the table to find the following:

x	-2	1	4	7	10
$F(x)$	2	-3	5	-1	0
$f(x)$	4	-2	7	11	-4
$f'(x)$	6	9	-8	-13	1
$f''(x)$	-5	17	3	-7	8

10.) $\int_{1}^{7} f(x)\,dx$

11.) $\int_{-2}^{1} f(x)dx$

12.) $\frac{1}{6}\int_{10}^{4} f(x)dx$

13.) $\int_{4}^{-2} f'(x)\,dx$

14.) $\int_{1}^{1} [f'(x)+3]dx$

15.) $\int_{7}^{10} f'(x)dx$

16.) $\int_{1}^{10} f''(x)\,dx$

17.) $\int_{-2}^{1} 2f''(x)dx$

18.) $\int_{7}^{-2} f''(x)dx$

19.) $\int_{7}^{1} -4f'(x)\,dx$

20.) $\int_{-2}^{-2} f(x)dx$

21.) $\int_{10}^{4} f''(x)dx$

t	1	3	6	7
$m(t)$	0.2	0.8	0.25	0.1
$M(t)$	0.1	0.7	1.5	2.2

22.) Overnight, snow begins falling heavily in Plugville. Due to warmer temperatures, the falling snow is melting once it lands on the ground. The rate that the snow is melting at time t, where t is hours since 12AM, is given by $m(t)$, measured in inches per hour. Selected values of $m(t)$ and $M(t)$, the antiderivative of $m(t)$ are given in the table above.

A.) How many total inches of snow melted between 1AM and 7AM?

B.) Approximate $m'(5)$. Include units of measure.

C.) Write an equation of the tangent line to $M(t)$ at $t = 3$. Use this tangent line to approximate $M(4)$.